

EX.NO:5 ADT STACK USING EXCEPTION HANDLING

PROGRAM:

```
import java.io.*;
interface Mystack
{
    void push();
    void pop();
    void display();
}
class Stack_array implements Mystack
{
    final int n = 5;
    int stack[] = new int[n];
    int top = -1;
    public void push()
    {
        try
        {
            BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
            if(top == n - 1)
            {
                System.out.println("Stack Overflow");
                return;
            }
            System.out.print("Enter the element: ");
            int ele = Integer.parseInt(br.readLine());
            stack[++top] = ele;
        }
        catch(Exception e)
        {
            System.out.println(e);
        }
    }
    public void pop()
    {
        if(top == -1)
        {
            System.out.println("Stack Underflow");
            return;
        }
        System.out.println("Popped Element: " + stack[top]);
        top--;
    }
    public void display()
    {
        if(top == -1)
        {
            System.out.println("Stack is Empty");
            return;
        }
        System.out.println("Stack Elements:");
    }
}
```

```

        for(int i = top; i >= 0; i--)
        {
            System.out.println(stack[i]);
        }
    }
}

public class Main
{
    public static void main(String args[]) throws Exception
    {
        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
        Stack_array stk = new Stack_array();
        int ch;
        do
        {
            System.out.println("\n*** STACK USING ARRAY ***");
            System.out.println("1. Push");
            System.out.println("2. Pop");
            System.out.println("3. Display");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");
            ch = Integer.parseInt(br.readLine());
            switch(ch)
            {
                case 1:
                    stk.push();
                    break;
                case 2:
                    stk.pop();
                    break;
                case 3:
                    stk.display();
                    break;
                case 4:
                    System.out.println("Program Ended");
                    break;
                default:
                    System.out.println("Invalid Choice");
            }
        } while(ch != 4);
    }
}

```

OUTPUT:

*** STACK USING ARRAY ***

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the element: 10

*** STACK USING ARRAY ***

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 1

Enter the element: 20

*** STACK USING ARRAY ***

1. Push
2. Pop
3. Display
4. Exit

Enter your choice: 3

Stack Elements:

20

10